Customer Shopping Behavior and Retail Analysis

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Abstract

This project analyzes customer shopping behavior using a retail dataset to identify patterns in purchasing behavior, demographic trends, and revenue generation. The study integrates **Python for data preprocessing, SQL for analytical querying, and Power BI for visualization**, forming a complete end-to-end analytics pipeline.

The analysis reveals key insights related to customer segmentation, subscription impact, product performance, and revenue distribution, providing actionable recommendations for business optimization.

1. Introduction

In the modern retail landscape, data-driven decision-making plays a crucial role in improving customer engagement and maximizing revenue. Understanding customer behavior enables businesses to tailor marketing strategies, optimize inventory, and enhance customer satisfaction.

This project aims to:

- Analyze customer purchasing patterns
- Identify high-value customer segments
- Evaluate the impact of subscriptions and discounts
- Provide data-driven business recommendations

2. Dataset Description

The dataset contains **3,900 customer records**, each with multiple attributes describing purchasing behavior and demographics.

Features Included:

- Customer ID
- Age, Gender
- Item Purchased, Category
- Purchase Amount (USD)
- Location, Size, Color
- Season
- Review Rating
- Subscription Status

3. Data Preprocessing (Python)

Data preprocessing was carried out using Python to ensure the dataset was clean, consistent, and suitable for analysis.

3.1 Data Inspection

Initial exploration using df.info() and df.describe() helped understand data types, distributions, and overall structure.

3.2 Handling Missing Values

Missing values in the *Review Rating* column were filled using the median rating for each category to maintain accuracy.

3.3 Data Cleaning

Column names were standardized to lowercase and spaces were replaced with underscores to ensure consistency.

3.4 Feature Engineering

New age group categories (Young Adult(18-26), Adult(26-46), Middle-aged(46-61), Senior(61-71) were created, and categorical frequency values were converted into numerical form for analysis.

3.5 Redundancy Removal

The overlap between discount_applied and promo_code_used was identified, and one redundant column was removed to improve efficiency.

4. Exploratory Data Analysis (Python)

EDA was performed to understand distributions and trends.

Key Observations:

- Purchase amounts are centered around a consistent average
- Majority of customers fall within Adult and Middle-aged groups
- Clothing category shows highest transaction frequency

5. Database Integration (MySQL)

The cleaned dataset was exported into a **MySQL database** using SQLAlchemy.

Technologies Used:

- SQLAlchemy
- PyMySQL
- MySQL

Process:

1. Created database: customer_behaviour
2. Established connection using SQLAlchemy
3. Loaded dataset into table: customer_data

6. Advanced Analysis Using SQL

To extract deeper insights from the dataset, structured SQL queries were executed on the MySQL database. These queries helped uncover patterns in customer behavior, revenue distribution, and product performance.

6.1 Revenue Contribution by Gender

The analysis revealed that male customers contribute a significantly higher portion of total revenue compared to female customers. This indicates a stronger purchasing tendency within this segment, suggesting an opportunity for targeted marketing strategies aimed at balancing gender-based engagement.

6.2 High-Spending Customers with Discounts

Several customers were identified who made above-average purchases even after applying discounts. This indicates that discounts do not necessarily reduce revenue but instead encourage higher spending behavior among certain customer segments.

6.3 Top Rated Products

Products such as Shorts, Handbags, and Dresses emerged as the highest-rated items, with ratings approaching 4.7–4.9. This reflects strong customer satisfaction and highlights these categories as key drivers of customer experience.

6.4 Shipping Type vs Purchase Behavior

The comparison between standard and express shipping showed minimal difference in average purchase value, both hovering around approximately $59. This suggests that shipping type has little influence on customer spending decisions.

6.5 Impact of Subscription on Spending

Subscribed customers demonstrated slightly higher average spending compared to non-subscribers. However, non-subscribers contributed a larger share of total revenue due to their higher population, indicating untapped potential in subscription conversion.

6.6 Discount Usage Across Product Categories

Certain product categories such as Sweaters, Jackets, and Pants exhibited higher discount usage rates, often exceeding 20%. This suggests that discounts are more effective in specific categories and can be strategically applied.

6.7 Customer Segmentation Distribution

The majority of the customer base consists of approximately 2,500 returning customers, followed by 1,000 loyal customers, While new customers account for around 400. This indicates strong customer retention but highlights the need for new customer acquisition strategies.

6.8 Category-wise Product Popularity

Within major categories such as Clothing and Footwear, specific items consistently dominated purchases. with Clothing alone contributing over $100K+ in revenue highlighting strong consumer preference and guiding inventory decisions.

6.9 Subscription Adoption Among Repeat Buyers

Among customers with more than five previous purchases, only 26.5% are subscribed, while 73.5% remain non-subscribers, indicating a significant opportunity to convert frequent buyers into long-term subscribers and increase customer lifetime value.

6.10 Revenue Distribution by Age Group

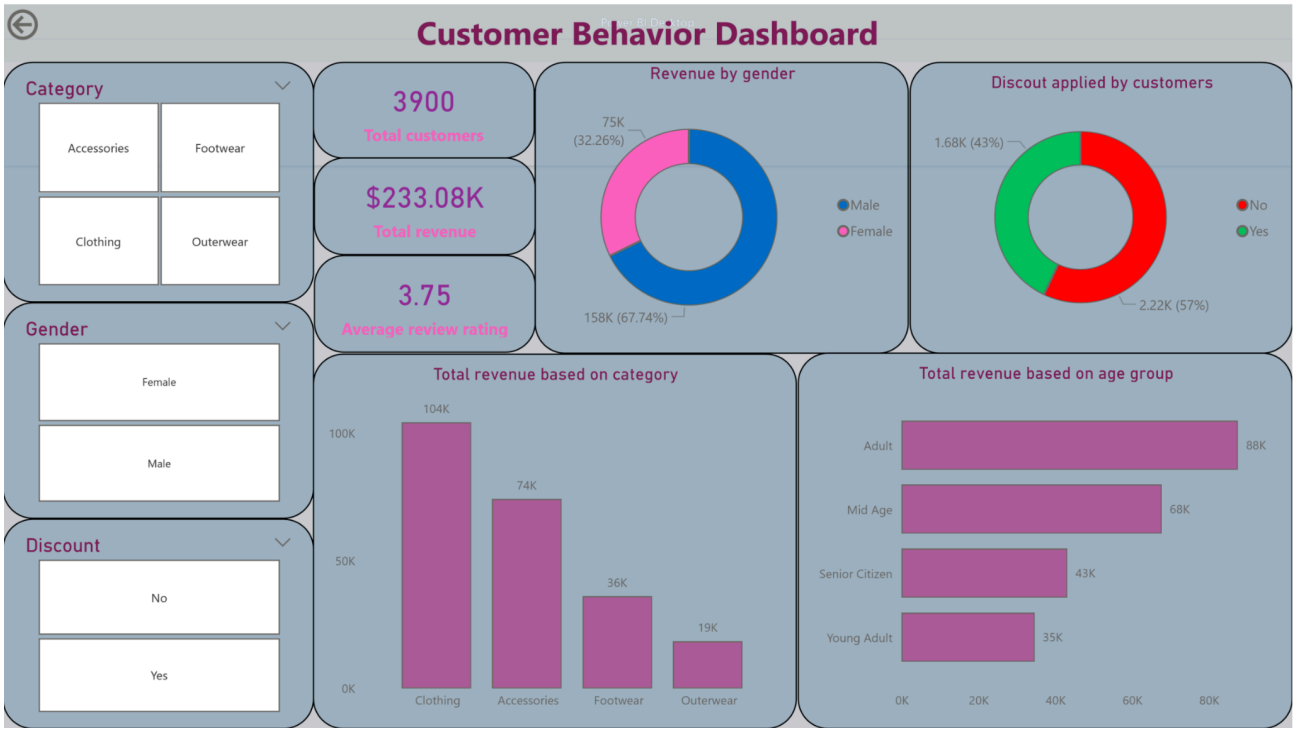
The Adult age group (26-46) emerged as the highest revenue-generating segment, followed by middle-aged customers. Younger and senior groups contributed less comparatively, indicating where marketing efforts should be concentrated.

7. Data Visualization (Power BI Dashboard)

An interactive dashboard was developed to visualize key metrics. Based on dashboard structure similar to reference :

Key Metrics:

- Total Customers: ~3,900
- Average Purchase: ~$59
- Average Rating: ~3.7



Dashboard Insights:

- **Subscription Distribution (Donut Chart)**

  A donut chart reveals that nearly **73%** of customers are non-subscribers, highlighting a strong opportunity to improve subscription adoption.

- **Gender Distribution (Donut Chart)**

  The gender split indicates that around **68%** of customers are male, contributing to a higher share of overall revenue.

- **Revenue by Category (Bar Chart)**

  A vertical bar chart shows Clothing as the dominant category with over **$100K** revenue, followed by Accessories, Footwear, and Outerwear.

- **Revenue by Age Group (Column Chart)**

  The Column chart highlights that the Adult (26-46) segment generates the highest revenue compared to other age groups.

- **Customer Segmentation Insights**

  The visuals indicate that returning customers form the majority, demonstrating strong retention within the business.

- **Interactive Filtering and Cross-Analysis**

  The dashboard enables dynamic filtering by category, gender, and subscription status, allowing deeper exploration of customer behavior.

8. Business Insights

The analysis indicates that customer behavior is primarily driven by age demographics and product preferences, with the 30–49 age group contributing the highest share of revenue and Clothing emerging as the leading category.

While returning customers form a strong base, subscription adoption remains low at around 27%, presenting a key opportunity for growth.

Additionally, discount strategies effectively drive engagement but should be optimized to maintain profitability.

To improve performance, the business can focus on targeted marketing for high-value segments, enhance subscription incentives to increase customer lifetime value, and strategically promote underperforming categories to achieve a more balanced revenue distribution.

9. Conclusion

This project successfully demonstrates a complete data analytics workflow by integrating:

- **Python for preprocessing and EDA**
- **SQL for advanced analysis**
- **Power BI for visualization**

The insights generated provide meaningful business value and can support strategic decision-making in retail environments.